

Raquel Melgar

PhD Student in Algebraic Combinatorics

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I'm currently a third year PhD student under the supervision of Jean-Christophe Aval, at LaBRI in Bordeaux (France). I am a member of the "Combinatorics and interactions" research group.

I am interested in studying algebraic structures that arise in combinatorial problems, such as graph coloring, with a particular focus on symmetric and quasisymmetric functions. I am also interested in numerical semigroups and problems involving integer points in rational polyhedra.

RESEARCH EXPERIENCE	PhD thesis , LaBRI Title: <i>Polynomial invariants of discrete structures: algebraic and combinatorial studies</i> . Supervised by Jean-Christophe Aval. Master thesis , Universidad de Valladolid Title: <i>Exponential expression of generating functions and their combinatorics</i> . Supervised by Antonio Campillo-López. Bachelor thesis in Mathematics , Universidad de Valladolid Title: <i>Polyhedra and convex sets. A generalization of volume</i> . Supervised by Antonio Campillo-López.	Feb 2024 - Present Jan 2023 - Jul 2023 Jan 2022 - Sep 2022
EDUCATION	Master in Mathematics (60 ECTS) , University of Valladolid Specialization: Combinatorial algebra, Geometric Algebraic Codes and Singularities. Bachelor in Mathematics (240 ECTS) , University of Valladolid.	2022-2023 2018-2022
TALKS	Journées annuelles du GT CombAlg , Calais (France). Talk: <i>On the combinatorial Hopf algebra of signed graphs</i> 7th GSMAAC , Bristol (UK). Talk: <i>On the combinatorial Hopf algebra of signed graphs</i> 95 Séminaire Lotharingien de Combinatoire , Villeurbanne (France). Journées de Combinatoire de Bordeaux 2026 , Talence (France). Talk: <i>Chromatic quasisymmetric functions for signed graphs</i> XXI Spanish Meeting on Computational Geometry , Santander (Spain). Talk: <i>Chromatic quasisymmetric functions for signed graphs</i> . Combinatorics and Interactions seminar , Talence (France). Talk (in French): <i>Fonctions chromatiques quasisymétriques pour les graphes signés</i> . Open Day of the Doctoral School for Mathematics and Computer Science Bordeaux (France). Poster: <i>Algebraic invariants associated to graph coloring</i> . JNIFM 2025: Journées Nationales du GDR IFM , Bordeaux (France). Poster: <i>Algebraic invariants associated to graph coloring</i> . Journées de Combinatoire de Bordeaux 2025 , Talence (France). RSME's VII Congress of Young Researchers , Bilbao (Spain). Talk: <i>Algebraic invariants associated to graph coloring</i> . Parallel Session AL05 Applied Nonlinear Algebra.	Jun2026 Apr2026 Mar2026 Feb2026 Jun 2025 Jun 2025 March 2025 March 2025 Feb2025 Jan 2025

	XVIII Annual Young Foreser Researchers Meeting , Palencia (Spain). Talk: <i>A voxel-based ray tracing model to simulate sunlight captured by trees</i>	Jan 2024
WORKSHOPS	Combinatorial Hopf algebras and modern Lie theory with applications. Nordfjordeid (Norway) Lecturers: Claudia Malvenuto, Frédéric Patras and Christophe Reutenauer. Santander Workshop on Geometric and Algebraic Combinatorics. University of Cantabria (Spain). Beyond Permutahedra and Associahedra , Luminy (France).	Jun 2024 Jan 2024 Dec2025
TEACHING (IN FRENCH)	IUT Bordeaux, Département GMP First semester. Mathematics TD and TP (48h). IUT Bordeaux, Département GMP Second semester. Notions sur les méthodes de calculs numériques TD (16h). IUT Bordeaux, Département GMP First semester. Mathematics TD and TP (48h). IUT Bordeaux, Département GMP Second semester. Notions sur les méthodes de calculs numériques TD (16h).	Fall 2025 Spring 2026 Fall 2024 Spring 2025
PUBLICATIONS	[1] J.C. AVAL, R. MELGAR, <i>Chromatic quasisymmetric functions for signed graphs</i> , Submitted: https://arxiv.org/abs/2508.20200 [2] A. CAMPILLO LÓPEZ, R. MELGAR, <i>Poincaré series of semigroups</i> , To appear in Contemporary Mathematics Proceedings Volume: 115AM Algebraic and topological interplay of algebraic varieties. [3] A. CAMPILLO LÓPEZ, R. MELGAR, <i>Analytic expansions and Poincaré series</i> , Ann.Polon.Math. 135 (2025), 79-103.	
COMPETENCES	Languages Spanish is my first language but I am also fluid in French and English. Some notions of Mandarin Chinese (certified HSK2 level). Techniques Python, Matlab, Sage, R.	